

★ LEVEL 1 — BASIC / THEORY TYPE ANSWERS

A. VERY SHORT ANSWER (1–2 lines) (Q1–40)

1. **Relation:** A relation from set A to set B is a subset of the Cartesian product $A \times B$, i.e., a set of ordered pairs (a, b) where $a \in A$ and $b \in B$.
2. **Domain:** The domain of a relation is the set of all first elements of the ordered pairs.
3. **Range:** The range of a relation is the set of all second elements of the ordered pairs.
4. **Codomain:** Codomain is the set in which all outputs (second elements) of a relation or function lie; may include elements not mapped.
5. **Reflexive Relation:** A relation R on set A is reflexive if $(a, a) \in R$ for all $a \in A$.
6. **Symmetric Relation:** R is symmetric if $(a, b) \in R \implies (b, a) \in R$.
7. **Transitive Relation:** R is transitive if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$.
8. **Equivalence Relation:** A relation that is reflexive, symmetric, and transitive.
9. **Partial Order Relation:** A relation that is reflexive, antisymmetric, and transitive.
10. **Identity Relation:** $I_A = \{(a, a) : a \in A\}$. Example: if $A = \{1, 2\}$, $I_A = \{(1, 1), (2, 2)\}$.
11. **Universal Relation:** $U = A \times A$, relates every element to every element. Example: $A = \{1, 2\}$, $U = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.
12. **Antisymmetric Relation:** If $(a, b) \in R$ and $(b, a) \in R \implies a = b$.
13. **Function:** A relation $f : A \rightarrow B$ where each $a \in A$ has exactly one image in B .
14. **Is every relation a function?** No, because a relation may assign more than one value to an element of the domain.
15. **Domain of a Function:** The set of all inputs (first elements) for which the function is defined.
16. **Range of a Function:** The set of all outputs (images) actually obtained by the function.
17. **Codomain of a Function:** The set into which the function maps its elements. Example: $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$, codomain = $\mathbb{R}$.
18. **One-One (Injective) Function:** Different inputs map to different outputs.
19. **Onto (Surjective) Function:** Every element of codomain is an image of some element of the domain.
20. **One-One Onto (Bijective) Function:** Function that is both injective and surjective.

21. **Many-One Function:** Two or more elements of domain map to the same element in codomain.
 22. **Into Function:** Function whose range is a proper subset of the codomain.
 23. **Constant Function:** All elements of domain map to the same element of codomain.
 24. **Identity Function:** $f(x) = x$ for all $x \in A$.
 25. **Polynomial Function:** Function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$.
 26. **Modulus Function:** $f(x) = |x|$, gives non-negative value of x .
 27. **Signum Function:** $\text{sgn}(x) = 1$ if $x > 0$, 0 if $x = 0$, -1 if $x < 0$.
 28. **Greatest Integer Function:** $f(x) = [x]$, largest integer $\leq x$.
 29. **Least Integer Function:** $f(x) = \lceil x \rceil$, smallest integer $\geq x$.
 30. **Logarithmic Function:** $f(x) = \log_a x$, inverse of exponential function.
 31. **Exponential Function:** $f(x) = a^x$, base $a > 0$, $a \neq 1$.
 32. **Invertible Function:** Function that has an inverse function.
 33. **Inverse Function:** f^{-1} such that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.
 34. **Composition of Functions:** $(g \circ f)(x) = g(f(x))$.
 35. **Invertible Condition:** A function is invertible if it is one-one and onto.
 36. **Mapping Diagram:** Diagram showing elements of domain mapped to codomain.
 37. **Vertical Line Test:** Graphical test to check if a curve is a function.
 38. **Horizontal Line Test:** Graphical test to check if a function is one-one (invertible).
 39. **Composite Relation:** If $R \subseteq A \times B$ and $S \subseteq B \times C$, composite $S \circ R = \{(a, c) : \exists b, (a, b) \in R, (b, c) \in S\}$.
 40. **All equivalence relations symmetric?** Yes, by definition.
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B. SHORT ANSWER (3–4 lines) (Q41–80)

41. **Reflexive but not symmetric:** $R = \{(1, 1), (2, 2), (1, 2)\}$ on $A = \{1, 2\}$.
42. **Symmetric but not transitive:** $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ on $A = \{1, 2, 3\}$.
43. **Transitive but not reflexive:** $R = \{(1, 2), (2, 3), (1, 3)\}$ on $A = \{1, 2, 3\}$.
44. **Universal relation reflexive:** $U = A \times A$ contains all (a, a) so reflexive.
45. **Equivalence relation in real life:** "Has same birthday as" among people.
46. **Antisymmetric example:** $R = \{(1, 1), (2, 2), (1, 2)\}$; no pair $(2, 1)$ exists.
47. **Number of relations from $A = \{1, 2\}$ to $B = \{3, 4, 5\}$:** $2^{2 \times 3} = 64$.

48. **Identity function is one-one and onto:** Each element maps to itself; all domain elements have unique image, all codomain elements covered.
49. **Many-one into function:** $f : \{1, 2, 3\} \rightarrow \{a, b\}$ with $f(1) = a, f(2) = a, f(3) = b$.
50. **Onto but not one-one:** $f : \{1, 2, 3\} \rightarrow \{a, b\}$ with $f(1) = a, f(2) = a, f(3) = b$.
51. **Invertible function example:** $f(x) = 2x + 3, f^{-1}(x) = (x - 3)/2$.
52. **Graphical representation:** Plot of domain vs range points.
53. **$\sqrt{x}$ not defined for negative x:** Square root of negative number is not real.
54. **Range of $y = |x|$:** $[0, \infty)$.
55. **Domain & Range of $y = 1/x$:** Domain: $x \neq 0$, Range: $y \neq 0$.
56. **Domain & Range of $y = \sqrt{4 - x^2}$:** Domain: $[-2, 2]$, Range: $[0, 2]$.
57. **$y = x^2$ one-one on $\mathbb{R}$?** No, fails horizontal line test; not invertible on all $\mathbb{R}$.
58. **$y = x^3$ invertible:** Yes, strictly increasing, one-one.
59. **Composition example:** $f(x) = x + 2, g(x) = 3x, (g \circ f)(x) = 3(x + 2) = 3x + 6$.
60. **Composition not commutative:** $f(x) = x + 2, g(x) = x^2, g(f(x)) = (x + 2)^2 \neq f(g(x)) = x^2 + 2$.
61. **Domain of $f(x) = \log(x - 2)$:** $x > 2$.
62. **Domain of $f(x) = \sqrt{x + 5}$:** $x \geq -5$.
63. **Domain of $f(x) = 1/(x^2 - 9)$:** $x \neq 3, -3$.
64. **$f(x) = |x|, f(-3) = 3$.**
65. **f, g one-one $\Rightarrow g \circ f$ one-one:** True; composition of injective functions is injective.
66. **$R = \{(1,2), (2,1), (1,1), (2,2)\}$ digraph:** Nodes 1,2 with arrows both ways and loops.
67. **Equivalence relations on 2-element set:** 2 possibilities: $\{(1, 1), (2, 2)\}, \{(1, 1), (2, 2), (1, 2), (2, 1)\}$.
68. **Partition from equivalence relation:** Divides set into disjoint equivalence classes.
69. **Parity relation on integers:** aRb if $a - b$ is even; divides integers into even/odd classes.
70. **Domain of $f(x) = \arctan x$:** All real numbers, $\mathbb{R}$.
71. **If f is onto:** Range = codomain.
72. **Many-one mapping example:** $f : \{1, 2, 3\} \rightarrow \{a, b\}, f(1) = f(2) = a, f(3) = b$.
73. **Graph shows one-one:** Horizontal line intersects at most once.
74. **Every function $\mathbb{R} \rightarrow \mathbb{R}$ invertible?** No, e.g., $f(x) = x^2$ not one-one.
75. **$f(x) = \sin x$ not invertible:** Not one-one over $\mathbb{R}$.

76. **Trigonometric functions invertible:** Restrict domain, e.g., $\sin x : [-\pi/2, \pi/2] \rightarrow [-1, 1]$.
77. **Domain** $f(x) = \sqrt{5 - 3x} : 5 - 3x \geq 0 \Rightarrow x \leq 5/3$.
78. **Range** $f(x) = x^2 + 4 : [4, \infty)$.
79. **$f(x)=|x|$ inverse?** No, not one-one.
80. **Algebra of functions:** Addition, subtraction, multiplication, division, and composition of functions.
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I can continue with **C. LONG ANSWER (Q81–120)** in the next message because the answer will be very long.

Do you want me to continue and give **all long answers 81–120 in full detail** next?

yes

Perfect! Let's continue with **C. LONG ANSWER (5–8 lines) (Q81–120)**. I'll provide **detailed explanations with examples and reasoning**.

C. LONG ANSWER (5–8 lines) (Q81–120)

81. Reflexive, symmetric, transitive relations with examples:

- **Reflexive:** Every element is related to itself. Example: $R = \{(1, 1), (2, 2), (3, 3)\}$.
- **Symmetric:** If aRb then bRa . Example: $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$.
- **Transitive:** If aRb and bRc then aRc . Example: $R = \{(1, 2), (2, 3), (1, 3)\}$.

82. "Is equal to" relation is equivalence:

- Let $R = \{(a, b) : a = b\}$ on set A .
- **Reflexive:** $a = a$. **Symmetric:** If $a = b$, then $b = a$. **Transitive:** If $a = b$ and $b = c$, then $a = c$.
- Hence it is an equivalence relation.

83. "Divides" relation on $\mathbb{N}$ is antisymmetric and transitive:

- Define aRb if a divides b .
- **Antisymmetric:** If $a|b$ and $b|a \Rightarrow a = b$.

- **Transitive:** If $a|b$ and $b|c \Rightarrow a|c$.
- Reflexive: $a|a$, so also a partial order.

84. Relation $R = \{(a,b): a-b \text{ even}\}$ is equivalence:

- **Reflexive:** $a-a=0$ even.
- **Symmetric:** If $a-b$ even $\Rightarrow b-a$ even.
- **Transitive:** If $a-b$ even and $b-c$ even $\Rightarrow a-c$ even.

85. Difference between equivalence and partial order:

- Equivalence: reflexive, symmetric, transitive; partitions set into classes.
- Partial order: reflexive, antisymmetric, transitive; defines hierarchy/order.

86. All possible relations on $A=\{1,2\}$:

- $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.
- Number of relations $= 2^4 = 16$. Examples: $\emptyset, \{(1,1)\}, \{(1,2),(2,1)\}, \dots$

87. Function can't assign two images to same element:

- Definition: Each element in domain has exactly **one image**.
- Assigning two images violates this, hence not a function.

88. $f(x) = 1/x$ is neither even nor odd:

- Check: $f(-x) = -1/x$.
- $f(-x) \neq f(x) \Rightarrow$ not even; $f(-x) \neq -f(x) \Rightarrow$ not odd.

89. Even and odd functions with graph:

- Even: symmetric about y-axis, e.g., $f(x) = x^2$.
- Odd: symmetric about origin, e.g., $f(x) = x^3$.
- Graph shows visual symmetry.

90. One-one onto function has unique inverse:

- Bijective ensures each y corresponds to exactly one x .
- Hence inverse is unique mapping from codomain to domain.

91. Composition examples:

1. $f(x)=x+1, g(x)=2x \rightarrow g \circ f=2(x+1)=2x+2$
2. $f(x)=x^2, g(x)=x+3 \rightarrow g \circ f=x^2+3$
3. $f(x)=\sin x, g(x)=\cos x \rightarrow g \circ f=\cos(\sin x)$

92. Composite of invertible functions is invertible:

- If f^{-1} and g^{-1} exist, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

93. $\sqrt{x^2} = |x|$:

- Square root always non-negative, hence $\sqrt{x^2} = x$ if $x \geq 0$, $-x$ if $x < 0$, which is $|x|$.

94. $f(x) = x^3 - 1$ invertible?

- $f'(x) = 3x^2 > 0 \Rightarrow$ strictly increasing.
- One-one and onto $\mathbb{R} \rightarrow \mathbb{R}$, hence invertible.

95. $f(x)=x^2$ is not one-one but becomes one-one on $x \geq 0$:

- On $\mathbb{R}$, $f(-2)=f(2)=4 \rightarrow$ not one-one.
- Restrict to $x \geq 0 \rightarrow$ strictly increasing $\rightarrow$ one-one $\rightarrow$ invertible.

96. Onto functions using mapping diagrams:

- Draw arrows from each domain element to codomain such that all codomain elements are covered.

97. Domain and range of $f(x)=1/\sqrt{4-x}$:

- $4-x > 0 \rightarrow x < 4$; denominator $\neq 0 \rightarrow x \neq 4$.
- Range: $(0, \infty)$

98. Codomain vs range significance:

- Codomain: set of possible outputs (may include elements not achieved).
- Range: actual outputs attained.

99. Number of possible functions from A to B:

- $|A|=m, |B|=n \rightarrow$ Number of functions $= n^m$.

100. All functions from $A=\{1,2\}$ to $B=\{3,4\}$:

- $f_1: 1 \rightarrow 3, 2 \rightarrow 3$; $f_2: 1 \rightarrow 3, 2 \rightarrow 4$; $f_3: 1 \rightarrow 4, 2 \rightarrow 3$; $f_4: 1 \rightarrow 4, 2 \rightarrow 4$.

101. Composition of two constant functions is constant:

- $f(x)=2, g(x)=5 \rightarrow g(f(x)) = g(2) = 5 \rightarrow$ constant.

102. Composition defined but reversed not:

- $f: \{1,2\} \rightarrow \{3,4\}, g: \{3\} \rightarrow \{5\} \rightarrow g \circ f$ undefined for $f(2)=4$, but $f \circ g$ defined.

103. If f invertible, then $f \circ f^{-1} = \text{identity}$:

- $f(f^{-1}(y)) = y$ for all y in codomain.

104. Identity function is invertible:

- $f(x) = x \rightarrow f^{-1}(x) = x \rightarrow f \circ f^{-1} = \text{identity}$.

105. Domain and range $f(x)=\sqrt{x^2+1}$:

- $x^2+1 \geq 0 \rightarrow$ Domain: $\mathbb{R}$.
- $\sqrt{x^2+1} \geq 1 \rightarrow$ Range: $[1, \infty)$

106. $f(x)=|x|$ not invertible:

- Not one-one: $f(2)=f(-2)=2 \rightarrow$ inverse not defined.

107. $f(x)=x^3$ invertible:

- Strictly increasing $\rightarrow$ one-one $\rightarrow f^{-1}(x)=x^{1/3}$.

108. Inverse of one-one onto is unique:

- Suppose two inverses exist $\rightarrow$ must map each element to same preimage $\rightarrow$ unique.

109. Division of real numbers and domain:

- $f(x)=a/x \rightarrow x \neq 0$ to avoid division by zero.

110. Domain of $f(x)=\sin^{-1}x$:

- $\sin^{-1}x$ defined for $x \in [-1,1]$.

111. Continuity of basic functions (conceptual):

- Polynomials, $\sin x$, $\cos x$, e^x continuous on $\mathbb{R}$; no jumps/discontinuities.

112. $f(x)=\tan x$ not defined at odd multiples of $\pi/2$:

- $\tan x = \sin x / \cos x \rightarrow \cos x = 0$ at $x = (2n+1)\pi/2 \rightarrow$ undefined.

113. Domain of $f(x)=\sqrt{2x-1}+1/(x^2-4)$:

- $2x-1 \geq 0 \rightarrow x \geq 1/2$
- $x^2-4 \neq 0 \rightarrow x \neq 2, -2$
- Domain: $[1/2, 2) \cup (2, \infty)$

114. $f(x)=\log x$ invertible:

- Strictly increasing $\rightarrow$ one-one $\rightarrow f^{-1}(x)=e^x$

115. Constant function cannot be invertible:

- Not one-one; multiple x map to same $y \rightarrow$ no inverse exists.

116. 5 examples of functions used in physics:

1. Displacement-time: $s(t)$
2. Velocity-time: $v(t)$
3. Acceleration-time: $a(t)$
4. Force-mass: $F(m)$
5. Pressure-volume: $P(V)$

117. Injective vs surjective comparison with diagrams:

- Injective: different domain elements map to different codomain elements.
- Surjective: all codomain elements covered by some domain element.

118. $R=\{(1,1),(2,2),(3,3)\}$ equivalence:

- Reflexive, symmetric, transitive $\rightarrow$ equivalence relation.

119. Relation as subset of Cartesian product:

- $R \subseteq A \times B$; elements are ordered pairs (a,b) showing correspondence between sets.

120. Relation may not be function but function is always relation:

- Function: each element of domain has exactly one image $\rightarrow$ is a relation.
- Relation: may assign multiple values $\rightarrow$ not necessarily a function.